

Questions and Answers

Topic: Dynamic Systems Development Method (DSDM)

1. Define Dynamic Systems Development Method (DSDM). Explain why DSDM is important in modern software engineering.

Answer:

Dynamic Systems Development Method (DSDM) is an Agile software development methodology that focuses on rapid delivery of software solutions through iterative development, continuous user involvement, and flexible requirement management.

DSDM was developed as a structured Rapid Application Development (RAD) approach. It helps organizations deliver software quickly while maintaining alignment with business objectives.

The main philosophy of DSDM is:

“Deliver what the business needs when it needs it.”

The importance of DSDM includes:

- Provides faster software development and delivery.
- Supports changing business requirements.
- Encourages continuous communication with customers.
- Reduces development risks through iterative improvement.
- Focuses on delivering maximum business value.

Therefore, DSDM is suitable for modern software projects where requirements frequently change and rapid delivery is required.

2. Compare Traditional Software Development and DSDM.

Answer:

Traditional software development and DSDM follow different approaches for handling project constraints such as requirements, time, and resources.

Traditional methods focus on detailed planning and fixed requirements, whereas DSDM emphasizes flexibility, rapid delivery, and continuous improvement.

Factor	Traditional Development	DSDM
Development Approach	Sequential approach	Iterative approach
Requirements	Fixed from beginning	Flexible and prioritized
Time	May vary	Fixed
Resources	May vary	Fixed
Functionality	Fixed functionality	Variable functionality
Customer Feedback	Limited involvement	Continuous involvement
Delivery	Complete product delivered at end	Incremental delivery
Risk Detection	Late detection	Early detection

DSDM provides more flexibility because requirements can be adjusted while keeping project time and resources under control.

3. Describe the DSDM process model with its major phases.

Answer:

The DSDM process model consists of several phases that guide software development from initial analysis to final implementation.

The major phases are:

(a) Feasibility Study

This phase determines whether DSDM is suitable for the project.

Main deliverables:

- Feasibility Report
- Feasibility Prototype
- Development Plan
- Risk Log

(b) Business Study

This phase analyzes business requirements and priorities.

Deliverables:

- Prioritized Requirements List
- Business Area Definition
- System Architecture Definition

(c) Functional Model Iteration

This phase focuses on creating functional prototypes.

Activities include:

- Identify prototype functionality.
- Create functional prototype.
- Review prototype with users.

(d) Design and Build Iteration

This phase converts prototypes into a working system.

Activities:

- Identify design requirements.
- Create design prototype.
- Perform testing and review.

(e) Implementation

The final system is delivered to users.

Activities:

- User approval.
- User training.
- System deployment.
- Business review.

4. Explain the 80/20 rule used in DSDM.

Answer:

The 80/20 rule is an important principle of DSDM flexibility.

The rule states that:

“80% of business value can be achieved by implementing the most important 20% of requirements.”

DSDM assumes that all requirements do not provide equal value. Therefore, the development team focuses on the most important requirements first.

Example:

For an Online Shopping System:

Important requirements:

- User registration.
- Product searching.
- Payment system.
- Order management.

Less important requirements:

- Theme customization.
- Recommendation features.
- Advanced animations.

Advantages:

- Faster delivery.
- Better priority management.
- Reduced development cost.
- Improved customer satisfaction.

5. Explain Timeboxing in DSDM and discuss its importance.

Answer:

Timeboxing is a DSDM technique where development activities are completed within a fixed period of time.

Instead of extending deadlines to complete all requirements, DSDM adjusts functionality according to priority.

Characteristics of timeboxing:

- Fixed start and end dates.
- Usually lasts between 2–4 weeks.
- Focuses on essential requirements first.
- Remaining features move to future iterations.

Importance:

- Controls project schedule.
- Improves resource management.
- Prevents uncontrolled development.
- Encourages faster decision making.
- Ensures regular software delivery.

6. Explain the MoSCoW prioritization technique in DSDM.

Answer:

MoSCoW is a requirement prioritization technique used in DSDM to decide which features should be developed first.

The four categories are:

Category	Meaning
Must Have	Critical requirements necessary for project success.
Should Have	Important features but not required immediately.
Could Have	Optional features that improve user experience.
Won't Have This Time	Features postponed for future releases.

MoSCoW helps teams deliver the most valuable functionality within fixed time and resources.

7. Why is prototyping important in DSDM?

Answer:

Prototyping is one of the main techniques used in DSDM. A prototype is an early working version of a system that allows users to test and provide feedback.

Importance of prototyping:

- Helps users validate requirements.
- Improves communication between developers and customers.
- Allows early detection of errors.
- Reduces misunderstanding about system functionality.
- Increases confidence that the correct solution will be delivered.

Through continuous prototype evaluation, DSDM ensures that the final software satisfies actual business requirements.

8. Discuss the advantages and disadvantages of DSDM.

Answer:

Advantages:

- Provides rapid software delivery.
- Supports changing requirements.
- Encourages user involvement.
- Reduces development risks.
- Improves software quality through feedback.

Disadvantages:

- Requires continuous customer participation.
- Needs experienced development teams.
- Not suitable for every type of project.
- Less documentation may create maintenance problems.

DSDM is most effective for projects where requirements evolve and quick delivery is important.